demo_clustering

June 8, 2026

[15]:

```python

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[16]:

```python
"""
Demonstrating my clustering wrapper based on Scikit-learn and cuML, with full␣
↪GPU support
"""
import pandas as pd
import warnings
from ac.ml.clustering import ClusterMaker
warnings.filterwarnings("ignore")
```

[17]:

```python
"""
Loading a smaller version of my dataset containing all Tweets / Truth posts of␣
↪the US Presidency (Obama to Trump 2).
For each message, sentiment analysis has already been performed to discover␣
↪which kind of emotion the presidents express.
Emotion values range from 0 to 1.
"""
presidential = pd.read_csv("./data/small_presidential_tweets.csv")
presidential.fillna(value=0, inplace=True)
presidential.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2837 entries, 0 to 2836
Data columns (total 37 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Unnamed: 0      2837 non-null   int64
 1   Author          2837 non-null   object
 2   Social          2837 non-null   object
 3   Id              2837 non-null   int64
 4   Content         2837 non-null   object
 5   Words           2837 non-null   object
 6   Date            2837 non-null   object
 7   Time            2837 non-null   object
 8   Intensity       2837 non-null   float64
 9   anger           2837 non-null   float64
 10  fear            2837 non-null   float64
```

```
 11  disgust          2837 non-null   float64
 12  sadness          2837 non-null   float64
 13  joy              2837 non-null   float64
 14  surprise         2837 non-null   float64
 15  disapproval      2837 non-null   float64
 16  optimism         2837 non-null   float64
 17  nervousness      2837 non-null   float64
 18  pride            2837 non-null   float64
 19  admiration       2837 non-null   float64
 20  disappointment   2837 non-null   float64
 21  neutral          2837 non-null   float64
 22  annoyance        2837 non-null   float64
 23  approval         2837 non-null   float64
 24  realization      2837 non-null   float64
 25  desire           2837 non-null   float64
 26  excitement       2837 non-null   float64
 27  embarrassment    2837 non-null   float64
 28  confusion        2837 non-null   float64
 29  caring           2837 non-null   float64
 30  gratitude        2837 non-null   float64
 31  love             2837 non-null   float64
 32  grief            2837 non-null   float64
 33  relief           2837 non-null   float64
 34  curiosity        2837 non-null   float64
 35  remorse          2837 non-null   float64
 36  amusement        2837 non-null   float64
dtypes: float64(29), int64(2), object(6)
memory usage: 820.2+ KB
```

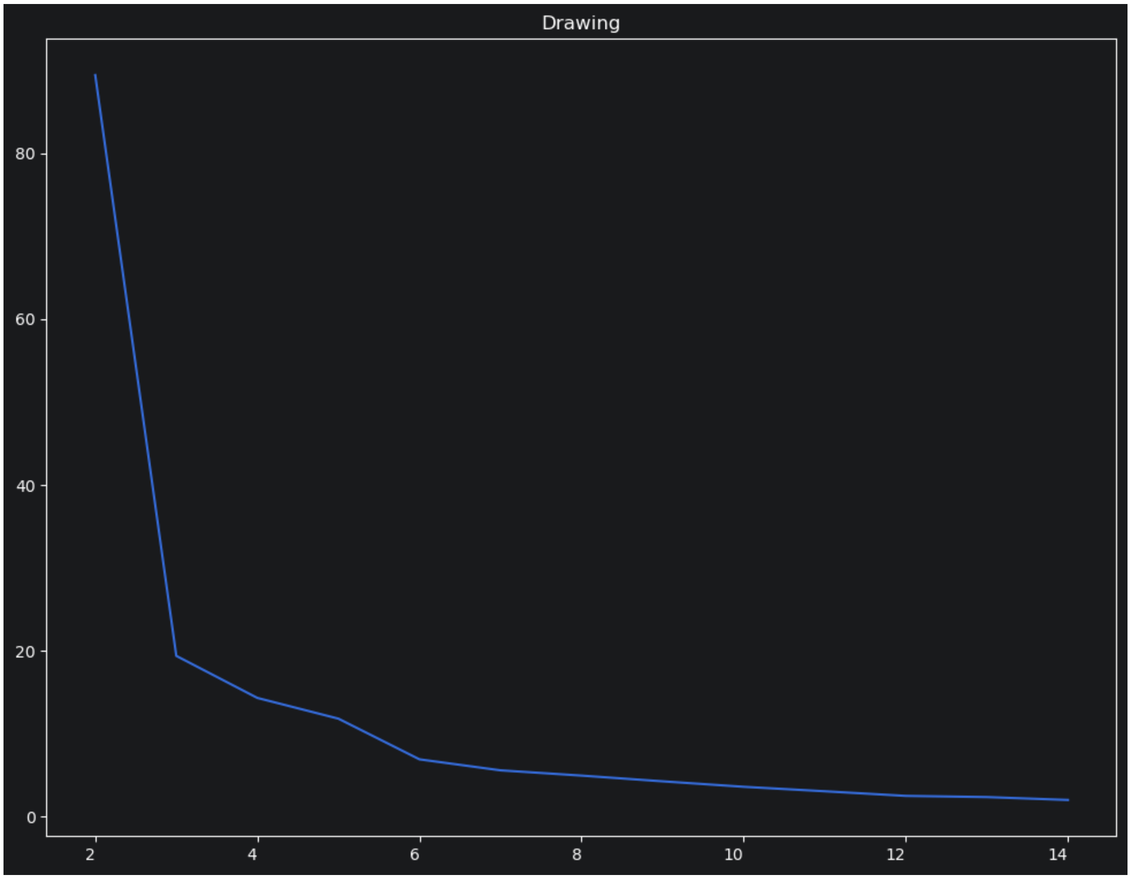
[18]:

```
"""
Let's pass the dataset to the ClusterMaker. We'll perform basic sentiment␣
↪analysis. How do posts expressing
"admiration" and "gratitude" relate to each other? We'll use a Kmeans algorithm
"""
cm = ClusterMaker(
    method="kmeans",
    data=presidential,
    subset=["admiration", "gratitude"],
    scaling="minmax",
    random="50%",
    cpu=False
)
```

[19]:

```
"""
Let's have a look to the SSE curve. It looks like the optimal n_clusters␣
↪hyperparameter amounts to 3...
"""
```

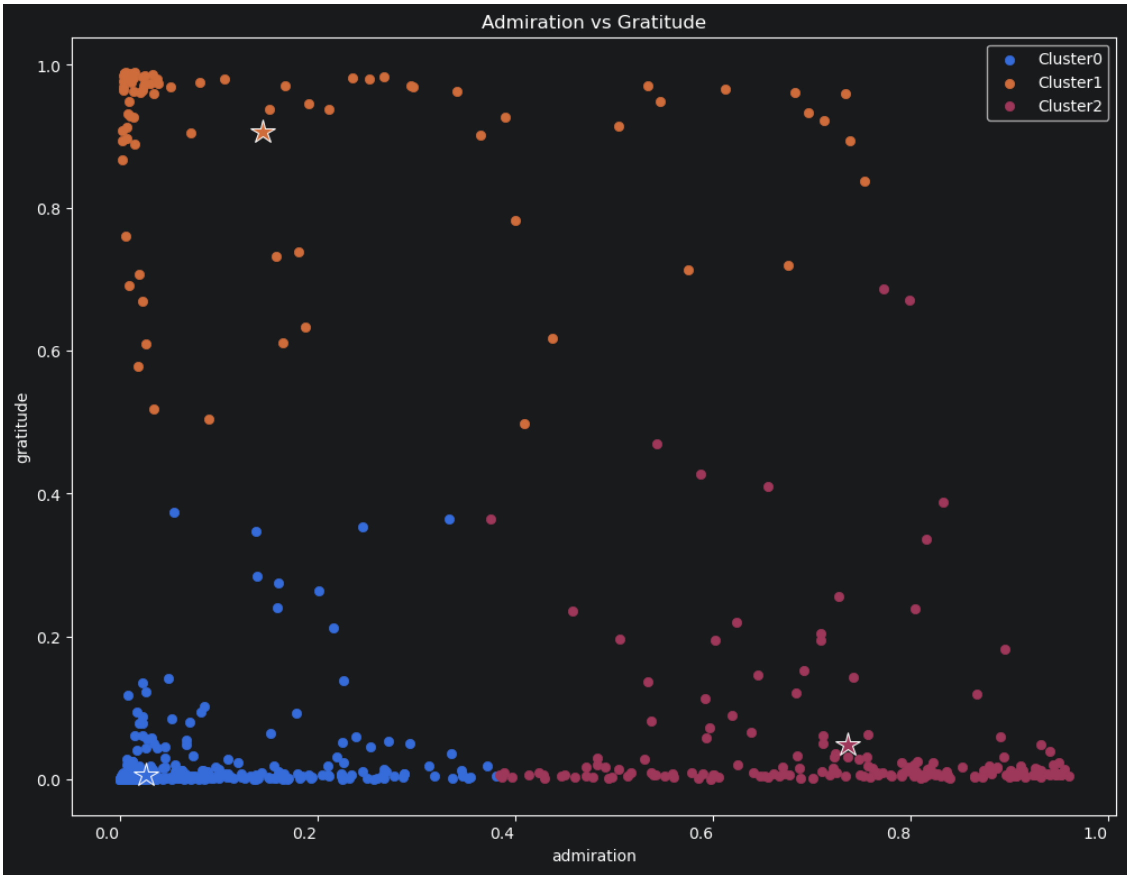
```python
cm.curve()
cm.draw(what="sse", show=True)
```



[19]: True

[20]:
```python
"""
We can let ClusterMaker choose the optimal hyperparameters automatically,␣
 ↪though. Just launch the explore() method.
Then, just fit the data and plot the clustering.
"""
cm.explore()
cm.fit()
cm.draw(
    what="clustering",
    x="admiration",
    y="gratitude",
    centroids=True,
    show=True,
    title="Admiration vs Gratitude",
    xlabel="admiration",
```

```
    ylabel="gratitude",
)
```



[20]: True